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1. Introduction

ABC Solutions Pvt Ltd is a mid-sized IT organization with approximately 250 employees operating across multiple departments. As the organization grows, the Human Resources (HR) department requires data-driven insights to improve workforce planning, talent retention, performance management, and diversity initiatives. Currently, HR-related data is scattered across multiple Excel files, making analysis time-consuming and inconsistent.

This project focuses on developing an HR Analytics Dashboard using Power BI, supported by a well-designed data warehouse model. The objective is to analyze employee data from the last three completed years to identify attrition patterns, performance trends, salary distribution, and workforce demographics. By applying Business Intelligence concepts such as star schema modeling, ETL/ELT processes, and DAX calculations, this dashboard enables management to make informed strategic HR decisions.

2. Part A – Theory

2.1 Difference between OLTP and OLAP

OLTP (Online Transaction Processing) and OLAP (Online Analytical Processing) systems serve different purposes in an organization. **OLTP systems** are designed to handle day-to-day business operations with a large number of short transactions, while **OLAP systems** are optimized for data analysis and reporting.

In an HR context, an **OLTP system** supports routine operational tasks such as employee onboarding, attendance marking, salary updates, leave applications, and performance score entry. For example, when an HR executive updates an employee's resignation date or records a new hire, the transaction must be processed quickly and accurately. OLTP databases are highly normalized, contain current data, prioritize fast insert, update, and delete operations. On the other hand, **OLAP systems** are used for analytical purposes and decision-making. In ABC Solutions Pvt Ltd, the HR Analytics Dashboard built in Power BI is an OLAP solution. It analyzes historical data across multiple years to identify trends such as attrition rate by department, salary growth over time, performance distribution, and workforce diversity.

OLAP systems use denormalized schemas (like star schema), store historical data, and support complex queries involving aggregations and calculations.

In summary, **OLTP supports daily HR operations**, while **OLAP supports strategic HR analysis and reporting**. Both systems complement each other, but OLAP is essential for data-driven HR decisions.

2.2 What is a Star Schema and Why Is It Used in BI?

The **Star Schema** is the most widely used architecture for data modeling in Business Intelligence (BI) and data warehousing. It is designed to optimize query performance, fast to process, and simple for business users to understand. In a Star Schema, data is organized into two distinct types of tables: **Facts** and **Dimensions**. When mapped out, the central Fact table connects to multiple Dimension tables, creating a shape that resembles a star.

The Fact Table (The "Center")

The Fact Table sits at the heart of the star. It contains the quantitative data—the "facts" or metrics you want to measure.

- *Examples:* salary amounts, performance scores, or attendance records.
  A FactSalary table may contain employee salary values linked to date and department.
- *Key Feature:* It usually contains foreign keys that link to the surrounding dimensions.

Dimension Tables (The "Points")

The Dimension Tables provide the context for the facts. They answer the "Who, What, Where, When, and Why" of the data.

- *Examples:* Date/Time, department, or employees' Demographics.
- The dimension tables provide descriptive context, such as DimEmployee, DimDepartment, and DimDate.
- *Key Feature:* These tables are "denormalized," meaning they are flat and simple, often containing descriptive text.

The Star Schema is preferred over traditional "normalized" databases (used in apps and websites) for several reasons:

- ***Query Performance:*** Because the data is denormalized, the database needs to perform fewer "joins." This leads to much faster response times when generating reports.
- ***Simplicity for Users:*** Business users and BI analysts can easily navigate the data. If you want to see "Salary" (Fact) by "Employee" (Dimension), the path is direct and obvious.
- ***Optimized for BI Tools:*** Modern tools like Power BI, Tableau, and Looker are specifically engineered to perform best when the underlying data is shaped like a star.
- ***Aggregations:*** It is highly efficient for "slicing and dicing," for example, viewing total Salary by year, then drilling down to the month, and then to a specific department.

Similarly, a Snowflake schema further breaks down (normalizes) the dimension tables into sub-dimensions. While Snowflake schemas save a bit of disk space, they increase complexity and slow down query performance. In modern BI, where storage is cheap but human time is expensive, the **Star Schema** is almost always the better choice.

2.3 Definitions

- ***Fact Table***

  a central table in a data warehouse that stores quantitative and measurable data related to business processes. In HR analytics, fact tables include metrics such as salary amounts, performance scores, bonus values, or attendance hours. Fact tables typically contain foreign keys that link to dimension tables and are primarily used for calculations, aggregations, and trend analysis.
- ***Dimension Table***

  This contains descriptive attributes that provide context to the facts. In HR systems, dimension tables include employee details (name, gender, age), department information, and date details. These tables allow users to filter, group, and slice data for meaningful analysis.
- ***Primary Key***

  A unique identifier that distinguishes each record in a table. For example, EmployeeID uniquely identifies each employee in the DimEmployee table. Primary keys ensure data integrity by preventing duplicate records.
- ***Foreign Key***

This is a column in a fact table that references the primary key of a related dimension table. For instance, EmployeeID in the FactSalary table acts as a foreign key linking salary data to employee details. Together, primary and foreign keys establish relationships that enable accurate and efficient HR reporting.

2.4 ETL vs ELT

ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) are two widely used approaches for integrating data into a data warehouse or analytical system. Both methods are used to prepare data for reporting and analysis, but they differ in the order and location of data transformation.

1. **ETL (Extract, Transform, Load)**

The Traditional "Gatekeeper" Approach

ETL is the traditional approach. Data is transformed *before* it reaches the destination. Think of it like a professional kitchen: ingredients are chopped, seasoned, and cooked before being served.

Data is extracted from the source, sent to a ***separate staging or processing server*** for transformation, and only then loaded into the data warehouse.

Suited for:

- ***Legacy systems****:* On-premise databases that can't handle heavy processing
- ***Strict security requirements****:* Sensitive data (e.g., credit card numbers or SSNs) can be masked or removed *before* storage
- ***Small or highly structured data****:* When the schema is well defined and unlikely to change.

2. ELT (Extract, Load, Transform)

The Modern "Cloud-Native" Approach

ELT is the modern, cloud-native approach. Raw data is loaded first, and transformations happen later using the power of the data warehouse. This is more like a buffet: all ingredients are available upfront, and each person customizes their plate as needed.

Data is extracted from the source and loaded directly into a data lake or cloud warehouse (such as Snowflake, BigQuery, or Redshift). Transformations are then performed inside the warehouse using SQL or similar tools.

Suited for:

- ***Big data****:* Massive datasets that would overwhelm a single transformation server
- ***Agility****:* New metrics or columns can be created without rebuilding the pipeline
- ***Unstructured data****:* Logs, JSON files, or semi-structured data that don't yet have a fixed schema

In HR, the decision often comes down to a trade-off between data privacy and analytical flexibility. HR data includes highly sensitive PII such as salaries, social security numbers, and medical records, raising the stakes significantly.

Choose ETL when privacy is non-negotiable

- **Regulatory compliance**: GDPR, HIPAA, and CCPA favor early masking and anonymization
- **Masking at the gate**: Sensitive fields are transformed before entering the warehouse
- **Legacy HRIS systems**: ETL offloads processing from older platforms
- **Payroll and tax accuracy**: Ensures a clean, single source of truth

Choose ELT for modern People Analytics

- **Exploratory analysis**: Raw data enables correlation studies and advanced analytics
- **Speed to insight**: New questions (e.g., D&I trends across regions) can be answered in hours, not weeks
- **Scalability**: Ideal for large organizations tracking engagement, collaboration tools, and behavioral data

2.5 Why Fact Tables Are Denormalized

In data warehousing and Business Intelligence systems, fact tables are intentionally denormalized to optimize query performance and support efficient analytical processing. Unlike Online Transaction Processing (OLTP) systems, which prioritize data integrity and frequent updates, Online Analytical Processing (OLAP) systems are designed primarily for data retrieval, aggregation, and reporting.

In an OLTP environment, normalization is used to minimize data redundancy and ensure consistency during insert, update, and delete operations. However, in a data warehouse, data is typically loaded periodically in batches and remains largely read-only. As a result, denormalization becomes advantageous because it significantly reduces the number of table joins required during analysis.

Fewer joins allow BI tools such as Power BI to execute complex queries faster, especially when dealing with large volumes of historical data.

In HR analytics, fact tables store measurable metrics such as salary amounts, performance scores, bonuses, and attrition indicators. These fact tables are linked to dimension tables like employee, department, and date. By deformalizing the structure, HR dashboards can quickly calculate metrics such as average salary by department, attrition rate by job role, or performance trends over time without excessive computational overhead.

Another benefit of denormalized fact tables is improved usability. The simplified star schema structure is easier for HR analysts and business users to understand, reducing the complexity of report creation and minimizing errors in analysis. Additionally, modern data warehouses and BI engines are optimized for scanning wide, denormalized tables, further improving aggregation performance.

Overall, denormalization of fact tables is essential for delivering fast, scalable, and user-friendly HR analytics solutions.

2.6 Steps to Create a Star Schema for HR Analytics

Creating a star schema for HR analytics involves a systematic process to ensure accurate reporting and optimal BI performance. The first step is to identify business requirements, such as analyzing employee attrition, salary trends, performance ratings, and workforce diversity. Understanding these requirements helps determine what data needs to be captured.

The second step is to define fact tables that store measurable data related to HR processes. For ABC Solutions Pvt Ltd, examples include FactSalary and FactPerformance, which contain numerical values such as salary amounts and performance scores. The third step is to identify dimension tables that provide descriptive context, such as DimEmployee, DimDepartment, and DimDate.

Next, primary keys are assigned to each dimension table, and corresponding foreign keys are added to the fact tables to establish relationships. These relationships are typically one-to-many, with dimensions filtering the fact tables.

Finally, the star schema is implemented and validated in Power BI. Relationships are tested, sample reports are created, and calculations are verified to ensure accuracy. Proper documentation is also maintained to support future scalability and model maintenance.

3. Data Model Diagram

The HR Analytics data model uses a **star schema** for easy analysis and fast performance.

- **DimEmployee**: central table with employee details such as EmployeeID, DepartmentID, DOB, Age, Employment Status, Recruitment and Resignation dates.
- **DimDepartment**: linked to DimEmployee to analyze data by department.
- **DimDate**: contains Date, Month, Month Number, and Year for time-based analysis.
- **FactSalaries**: stores salary amounts and dates.
- **FactPerformance**: stores performance scores and year.

All relationships are **one-to-many**, single direction, ensuring accurate filtering and calculations.

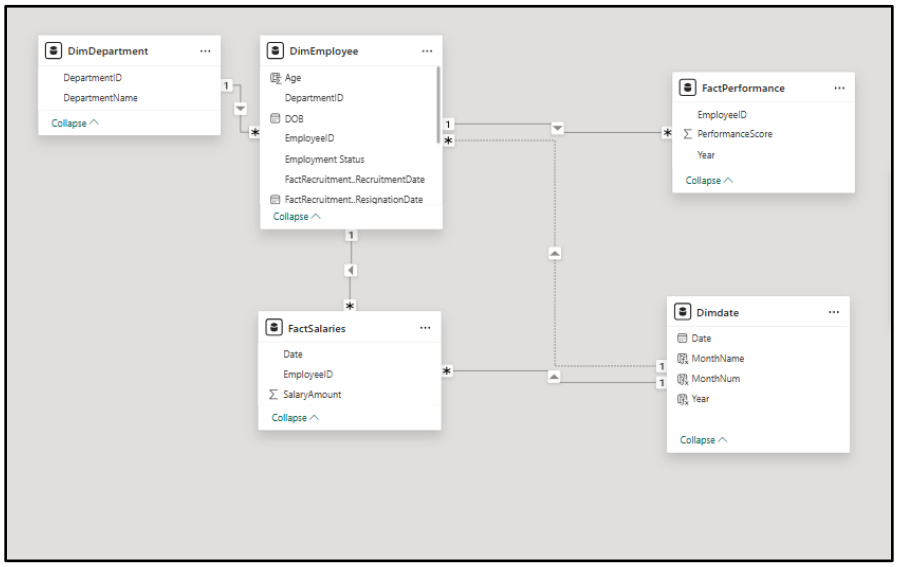


Figure 1

4 . DAX Formulas

DAX is used to create calculated columns and measures for HR analysis:

Calculated Columns:

- **Age** = difference between DOB and today.

  Age = DATEDIFF(DimEmployee[DOB],TODAY(),YEAR)

- **Years of Service** = difference between recruitment date and today.

  Years of Service = DATEDIFF (DimEmployee[FactRecruitment..RecruitmentDate],TODAY(),YEAR)

- **Service Category** = “New” if <=2 years, “Experienced” if >=3 years.

  Service Category = IF(DimEmployee[Years of Service]<= 2,"New", "Experienced")

- **Employment Status** = “Active” if Resignation Date is blank, else “Resigned”.

  = Table.AddColumn(#"Expanded FactRecruitment.", "Employment Status", each if [#"FactRecruitment..ResignationDate"] = null then "Active" else "Resigned")

Measures:

- **Employee**:
  - Total Employees = COUNTROWS (DimEmployee)
  - Active Employees = CALCULATE (COUNTROWS (DimEmployee), DimEmployee[Employment Status] = "Active")
  - Resigned Employees = CALCULATE (COUNTROWS (DimEmployee), DimEmployee[Employment Status]= "Resigned")
  - Attrition Rate = DIVIDE ([Resigned Employees],[Total Employees])

- **Salary**:
  - Total Salary = SUM (FactSalaries[SalaryAmount])
  - Average Salary = AVERAGE (FactSalaries[SalaryAmount])

- **Performance**:
  - Average Performance Score = AVERAGE(FactPerformance[PerformanceScore])
  - High Performers 2024 = CALCULATE(COUNTROWS(FactPerformance), FactPerformance[PerformanceScore] > 4, FactPerformance[Year] = "2024")
- **Diversity**:
  - Male % = DIVIDE (CALCULATE (COUNTROWS (DimEmployee),DimEmployee[Gender] = "Male"),[Total Employees])
  - Female % = DIVIDE (CALCULATE (COUNTROWS (DimEmployee),DimEmployee[Gender] = "Female"),[Total Employees])

- **Experience**:
    - Average Years of Service = AVERAGE (DimEmployee[Years of Service])
    - New vs Experienced Ratio % = DIVIDE(CALCULATE([Total Employees], DimEmployee[Service Category] = "New"),CALCULATE([Total Employees], DimEmployee[Service Category] = "Experienced"))

These formulas allow interactive dashboards with dynamic filtering.

5. Dashboard Screenshots

The dashboard shows key HR metrics using **KPI cards, charts, and filters**:

- **KPIs**: Total Employees, Active, Resigned, Attrition Rate, Average Salary, Avg. Performance Score, High Performers, New vs Experienced Ratio.
- **Charts**: Gender distribution, Department & Service Category, Salary trends by month/year, Attrition by department, Job roles with high attrition.
- **Filters**: Gender, Department, Service Category, Employment Status, Job Role.

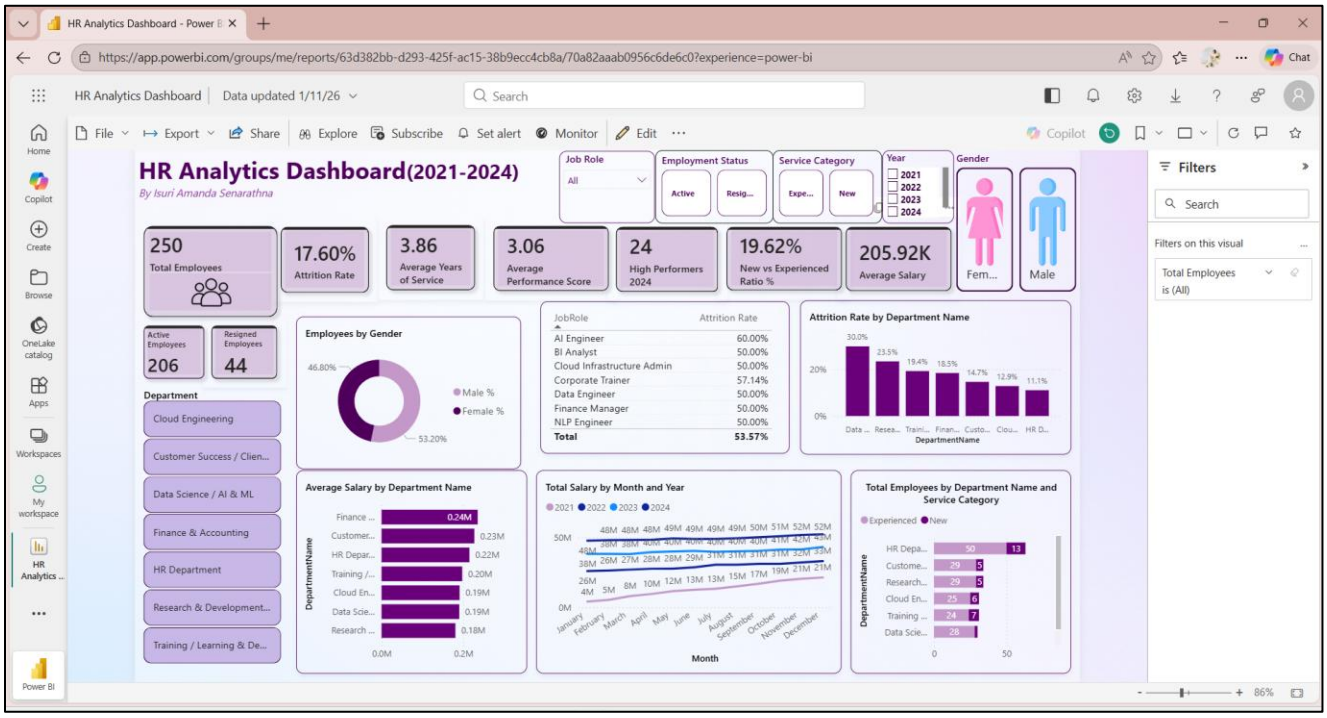


Figure 2

6. Part F – Insights

1. The HR Analytics Dashboard reveals that the **Data Science / AI & ML department** has the **highest attrition rate at 30%**, which is significantly higher than other departments. This indicates potential challenges related to workload pressure, role expectations, or job satisfaction within this department.

   At the job role level, **AI Engineers** experience the highest number of resignations. This trend suggests early-career retention issues, possibly due to high market demand for AI professionals, skill mismatches, or limited career progression opportunities. Management should consider targeted retention strategies such as competitive compensation, skill development programs, and clear growth paths for AI roles.

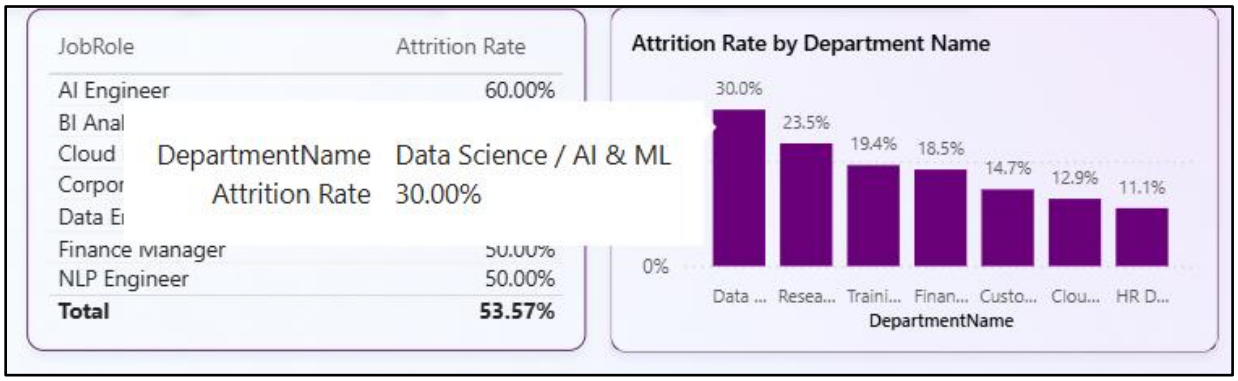


Figure 3

2. Industry **benchmarks** show that overall attrition rates in the IT sector and corporate India commonly fall in the **mid-teens range**. According to the *Deloitte India Talent Outlook Survey*, the average attrition rate across India Inc was **17.4% in 2024**, with the IT sector reporting an attrition rate of **15.1%** in the same period. This indicates that ABC Solutions' ***17.60%*** is **within a typical industry range**, not unusually high compared to broader IT and services benchmarks [1].

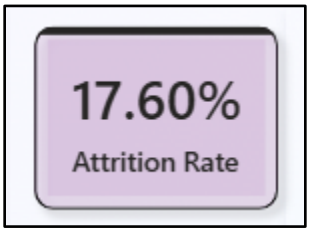


Figure 4

3. Analysis of the employee demographic data shows noticeable variation in male and female representation across departments at ABC Solutions Pvt Ltd.

   Customer Success / Client Relations and Finance & Accounting have a **higher proportion of female employees**, with females accounting for **52.94%** and **51.85%,** respectively. This indicates a relatively strong gender balance and inclusivity in these business and support-oriented departments.

   In contrast, several technical and specialized departments show a **male-dominated workforce**. Research & Development (R&D) has **52.94% male employees**, Data Science / AI & ML has **53.33% males**, Cloud Engineering records **54.84% males**, and Training / Learning & Development shows the highest male dominance at **61.29%**. The HR Department also has a higher male representation at **53.97%**.

   These findings indicate that **technical and engineering-focused departments tend to have lower female representation**, while customer-facing and finance roles demonstrate better gender balance. From a diversity and inclusion planning perspective, ABC Solutions should consider targeted initiatives such as inclusive recruitment strategies, mentorship programs, and career development opportunities aimed at increasing female participation in technical roles.

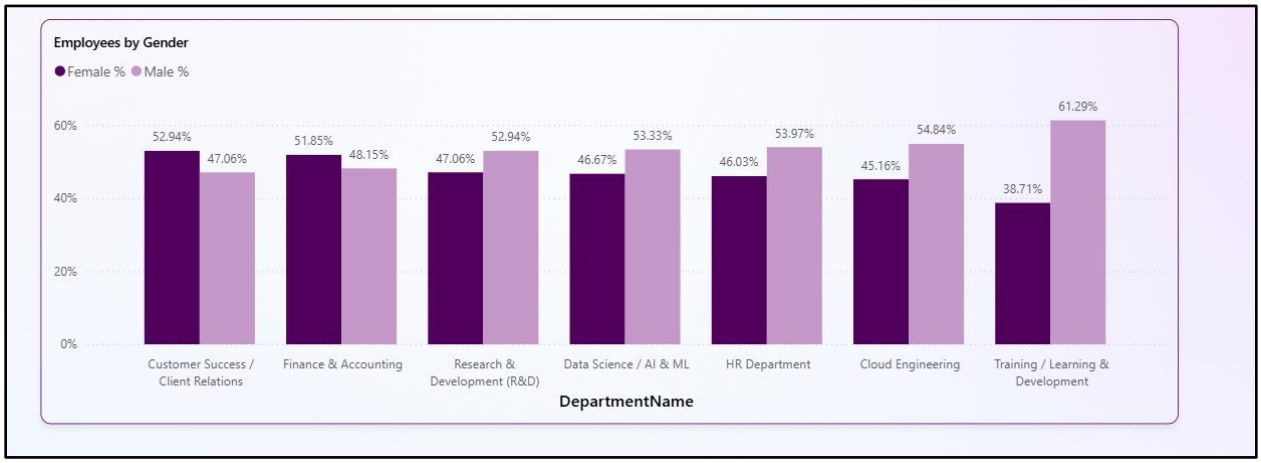


Figure 5

4. The average years of service at ABC Solutions Pvt Ltd is 3.86 years, which suggests **moderate workforce stability**, meaning that ABC Solutions is able to retain employees long enough to build experience and organizational knowledge, while still bringing in new talent to support growth and innovation. This level of tenure is beneficial for maintaining productivity, reducing excessive hiring costs, and ensuring continuity in project delivery.



Figure 6

5. Based on the analysis of attrition, turnover, gender distribution, and employee tenure, the HR management practices at **ABC Solutions Pvt Ltd** can be considered **partially effective**, but there is clear room for improvement.

The overall **attrition rate of 17.60%** shows that the company is able to retain most of its employees, which indicates that basic HR practices such as recruitment, salary management,

and employee administration are working reasonably well. With **206 out of 250 employees still active**, HR has maintained general workforce stability.

However, when looking deeper, **high attrition in certain departments** is a concern. Technical departments like **Data Science / AI & ML (30%)** and **Research & Development (23.5%)** have much higher turnover compared to departments like HR, which has the lowest attrition. This suggests that HR practices are less effective in retaining employees in highly competitive and skill-intensive roles. According to HR theory, specialized employees require stronger retention strategies, such as career growth.

7. Conclusion

The HR Analytics Dashboard developed for ABC Solutions Pvt Ltd demonstrates the practical application of Business Intelligence concepts to human resource data. By implementing a star schema data model with denormalized fact tables and clearly defined dimension tables, the system provides accurate, fast, and scalable reporting capabilities. The dashboard integrates key HR metrics such as employee count, attrition rate, salary trends, performance scores, and diversity measures, enabling data-driven decision-making.

The insights generated reveal that while the company maintains reasonable workforce stability overall, certain technical departments experience higher attrition rates. Gender distribution analysis indicates a need for targeted diversity initiatives in male-dominated departments. The average years of service suggest moderate employee retention, highlighting areas where HR policies can be refined. Overall, the dashboard supports strategic HR management by identifying critical areas for talent retention, workforce planning, and performance improvement.

References

[1] S. Basu, "India Inc's attrition rates in 2024 hit lowest point since 2020 amid subdued hiring environment, market un," The Economic Times, 6 03 2025. [Online]. Available: https://economictimes.indiatimes.com/jobs/mid-career/india-incs-attrition-rates-hit-lowest-point-since-2020-amid-hiring-slowdown/articleshow/118767184.cms?from=mdr&utm. [Accessed 6 01 2026].